Property 5:

$$^nC_r = \frac{n}{r} \times^{(n-1)} C_{(r-1)}.$$

Proof.

$$\begin{aligned}\frac{n}{r} \times^{(n-1)} C_{(r-1)} &= \frac{n}{r}\frac{(n-1)!}{(r-1)! \times ((n-1)-(r-1))!} \\ &= \frac{n(n-1)!}{r(r-1)!(n-r)!} = ^n C_r\end{aligned}$$

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Example 4.44 Evaluate the following: (i) $^{10}C_3$ (ii) $^{15}C_{13}$ (iii) $^{100}C_{99}$ (iv) $^{50}C_{50}$.

Solution:

(i)
$$\begin{aligned}^{10}C_3 &= \frac{10!}{7! \times 3!} = \frac{10 \times 9 \times 8 \times 7!}{7! \times 3 \times 2 \times 1} \\ &= \frac{10 \times 9 \times 8}{3 \times 2 \times 1} = 120\end{aligned}$$

(ii)
$$\begin{aligned}^{15}C_{13} &= \frac{15!}{2! \times 13!} = \frac{15 \times 14 \times 13!}{2 \times 1 \times 13!} \\ &= \frac{15 \times 14}{2 \times 1} = 105\end{aligned}$$

(iii)
$$^{100}C_{99} = \frac{100 \times 99!}{99!} = 100$$

(iv)
$$^{50}C_{50} = \frac{50!}{50! \times 0!} = 1.$$

Example 4.45 Find the value of $^5C_2$ and $^7C_3$ using the property 5

Solution:

$$^nC_r = \frac{n}{r} \times {}^{n-1}C_{r-1}.$$

Substituting $n = 5$ and $r = 2$, we get

$$\begin{aligned}^5C_2 &= \frac{5}{2} \times {}^4C_1 = \frac{5}{2} \times \frac{4}{1} \\ &= \frac{5 \times 4}{2 \times 1} = 10.\end{aligned}$$

Substituting $n = 7$ and $r = 3$, we get

$$\begin{aligned}^7C_3 &= \frac{7}{3} \times {}^6C_2 = \frac{7}{3} \times \frac{6}{2} \times {}^5C_{2-1} \\ &= \frac{7}{3} \times \frac{6}{2} \times {}^5C_1 = \frac{7}{3} \times \frac{6}{2} \times \frac{5}{1} \\ &= \frac{7 \times 6 \times 5}{3 \times 2 \times 1} = 35\end{aligned}$$